



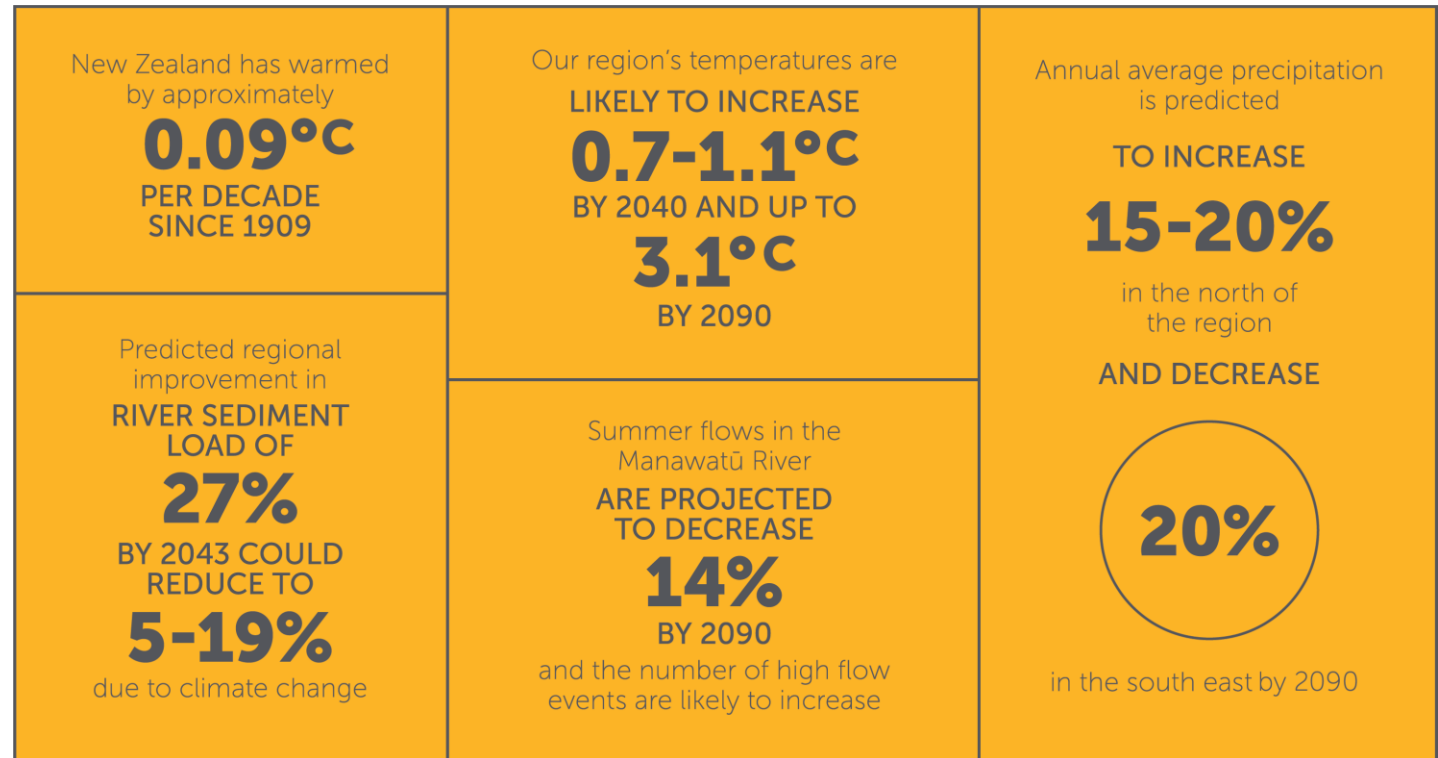
Feeling the Heat

Predicting and Mitigating River Temperatures Under Climate Change

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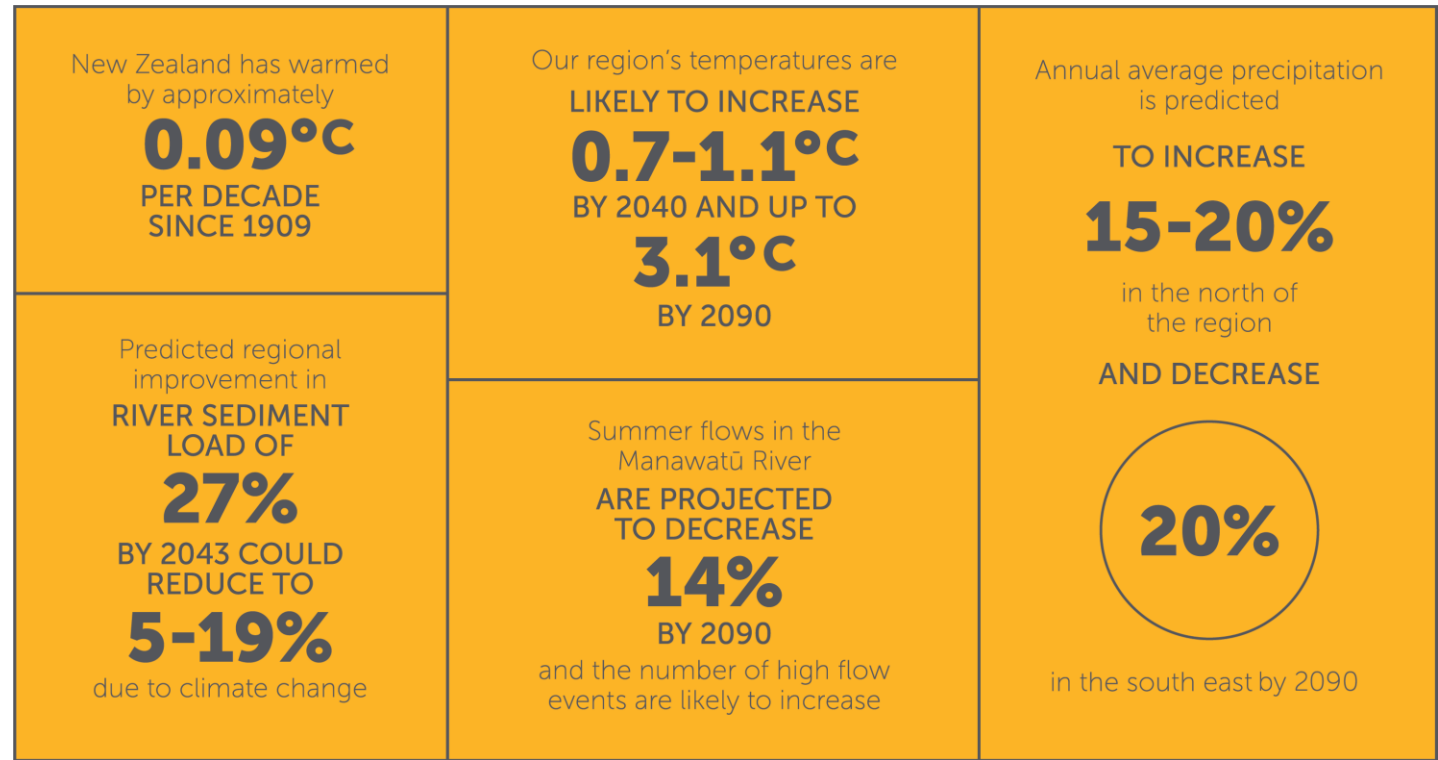
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- The Regional Council science team has a responsibility to consider climate change in policy development and actions.



Science

- Use expert opinion to understand the impact of climate change on freshwater quantity and quality, biodiversity and biosecurity, coastal and marine areas and natural hazards.
- Collect reliable environmental data, conduct research to inform policy development and actions.
- Provide monitoring information to the community and other agencies.

Other (non-related) climate change work

- Sediment in rivers

Period	Attribute band	% of REC2 segments by length achieving	
		Without climate-change	With climate-change
Mid-century (2040)	Band A	40%	23 - 32%
	Band B	62%	24 - 36%
	National bottom line	76%	25 - 38%
Late-century (2090)	Band A	40%	21 - 33%
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- How **river flows** will be affected by climate change (ongoing)
- Downscaling climate projections to regional scale (ongoing)

Specific interest – effect on freshwater ecosystems

Rising water temperatures threaten aquatic species, including macroinvertebrates and fish, which are already highly vulnerable to shifting climate conditions and loss of habitat.

Increased water temperature favours the proliferation of algae and toxic benthic cyanobacteria, especially when river flows drop.

This poses potential lethal risks to animals and significant health hazards to animals and humans.



Freshwater ecosystems

“Low vegetative areas... are more highly exposed due to reduced shade and altered natural flow patterns”

“Some native freshwater species... are sensitive to increases in water temperature”

Manawatū-Whanganui Regional Climate Change Risk Assessment Version 1.0 published 02 September 2021

Table 5.5: Risks to freshwater ecosystems from differing hazard

Hazard	Present	2050	2100	Commentary
Higher temperatures	Moderate	High	Extreme	Most freshwater species sensitive/intolerant of increases in temperature, with water temperature expected to increase with air temperature.
Change in rainfall	Low	High	High	Changes in rainfall expected to alter river/lake water balances - putting stress on freshwater ecosystems.
Drought	Low	High	High	Changes in rainfall expected to alter river/lake water balances - putting stress on freshwater ecosystems.

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Riparian shading: Blocking solar radiation.
Flow management: Augmenting river discharge.

Data and outcomes:

What we can provide:

- Monthly periphyton (algae) data at sites throughout the region.
- Annual Rapid Habitat Assessment at some sites (vegetation cover, etc)
- Flow data
- Water temperature
- Some high-level information about water allocated in catchments.

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What we hope for from MINZ (some/all):

- Ideas and approaches to get us started to answer these questions.
- Advice on what additional information is needed.
- Advice on defensible approaches to model, predict, mitigate river temperature rise.
- Guidance on uncertainty for any model/approach.



Questions?





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